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Title

PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

REVIEW REPORT – I

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Abstract

UAV systems for medical and emergency last-mile delivery must balance objectives that conflict: shorter routes carry higher flight risk, low-risk corridors drain more battery, and energy-efficient paths are rarely the fastest. No single fixed trade-off serves all payload types, yet existing planners apply the same objective weights to every mission regardless of what is being carried.

Existing multi-objective planners, including entropy-based, diversity-driven, and crowding-distance Pareto selection methods, encode objective weights as a fixed scalar before path search begins. Payload properties such as fragility class, payload weight class, and delivery urgency have no influence over which Pareto-optimal route is chosen. When the payload changes, operators must re-tune weights manually and re-run the planner, which is not workable under field conditions.

This work proposes a Payload-Aware Two-Tier UAV Route Planning system that separates Pareto front construction from route selection, introducing an explicit algebraic bridge between Tier 1 routing objectives and Tier 2 payload metadata.

Tier 1 runs a multi-objective A*/Dijkstra graph-search over a delivery airspace graph sourced from OpenStreetMap, producing a Pareto-optimal route set across distance, energy consumption, and flight risk. Tier 2 reads four operator-assigned payload tags and derives objective weights through a fixed formula: fragility class scales w_{risk}, delivery urgency scales w_{distance}, and payload weight class scales w_{energy}. These weights score every Pareto point through a utility function; the argmax is selected as the mission route. The system will be evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes) against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, and Tier 2 selection accuracy as metrics. Hardware validation will use a Pixhawk-equipped quadcopter through ArduPilot SITL simulation and outdoor flight tests.

Planned experiments will quantify Tier 1 improvements in hypervolume and Pareto spread over both baselines and will determine whether metadata-derived utility scoring selects non-extreme Pareto points for mixed-priority payloads that neither A* nor MOPSO can produce.

The principal contribution is a formula-driven mapping from payload metadata to routing objective weights, connecting mission context directly to Pareto point selection without expert intervention. This replaces context-blind diversity criteria with a direct algebraic link from payload description to route choice, and the Pareto front is refreshed on a periodic cadence rather than rebuilt for every mission, and is reused across all mission types without re-running the planner between deliveries.

1. Introduction

Unmanned aerial vehicles (UAVs) are a useful platform for last-mile delivery in scenarios where ground transport is impractical: medical supply chains to remote clinics, emergency drug drops during disaster response, and fragile diagnostic sample transport where road vibration risks contamination. In each scenario the UAV must fly a route from a dispatch hub to a delivery point across an airspace that contains obstacles, regulated no-fly zones, and variable wind corridors. A good route balances three conflicting physical objectives simultaneously: minimising total flight distance, minimising battery energy expenditure, and avoiding high-risk segments.

This three-way conflict is exactly what multi-objective UAV path planning is built to handle. Rather than collapsing distance, energy, and risk into one score before the search runs, a multi-objective planner keeps all three objectives separate through the search and returns a Pareto-optimal front: a set of routes where none can be improved on one objective without giving up ground on another [1]. The front tells the operator what trade-offs are available; it does not, by itself, say which route to fly.

One common way to sidestep the trade-off question is to form a weighted sum of the three objectives into a single scalar cost function and then run a conventional path planner such as A* on it. This scalar collapses the multi-objective problem into a single-objective one, and the operator adjusts the three weights to express the mission's priority before each flight. The approach works for a fixed mission type, but it fails immediately when the payload changes: emergency medicine requires a completely different weight profile from fragile biological samples or disaster-relief bulk supplies [2]. Manually re-tuning and re-running the planner between missions is operationally impractical in time-critical field conditions.

1.1 Background

Building the Pareto front, as introduced in Chapter 1, is well studied: the field has produced graph-search and population-based methods that generate well-spread, non-dominated route sets efficiently. The question that decides whether any such system is useful in practice is how the final route gets picked from the front. Chapter 2 reviews three directly relevant papers and finds the same answer in each: delivery context plays no role. None of them let what the UAV is carrying influence route choice, or even ask which route should be chosen for a given payload. That is the gap this project addresses.

Autonomous UAV systems for medical and emergency last-mile delivery operate

under a structural constraint that distinguishes them from general routing problems: the optimal route trade-off changes with what is being carried. A fragile diagnostic sample demands a low-vibration, low-turbulence corridor even at the cost of a much longer flight time. Emergency insulin requires the shortest path regardless of risk. Bulk disaster-relief supplies demand the most energy-efficient route to maximise delivery range.

Payload-awareness itself is not absent from the UAV literature: variable payload mass has been used to adapt the flight-speed and energy-consumption *model itself* during trajectory planning, recomputing the underlying physical equations as mass changes across a multi-drop mission [3]. What is absent is any use of payload metadata to *select* a point on a pre-computed multi-objective Pareto front. Tier 2's payload weight tag is a discrete, ordinal class (light/medium/heavy) that is one of four metadata tags algebraically blended into a static selection weight once per mission; it never alters the per-edge energy model or triggers re-planning, unlike [3]'s continuous mass-driven physics adaptation. Closing that gap—metadata-driven *selection* among a fixed front, not continuous model adaptation—is precisely the aim of Tier 2.

1.2 Motivation

The motivation for this work arises from three real operational scenarios that impose fundamentally incompatible routing requirements:

Fragile medical samples (tissue biopsies, blood samples, pathology specimens) are highly sensitive to vibration, g-forces, and thermal exposure. The dominant routing priority is minimising flight risk, defined here as exposure to turbulent corridors, sudden altitude changes, and wind-exposed segments. A route that adds 40% extra distance but halves the risk of vibration damage is operationally preferable.

Emergency medicine (insulin, epinephrine, antivenom) has a strict time window. The dominant routing priority is minimising route distance to minimise transit time. Flight risk and energy are secondary. For a payload with a 15-minute delivery window, the shortest path wins unconditionally.

Disaster-relief bulk supplies (water purification tablets, bandages, ration packs) weigh significantly more than medical payloads and are not time-critical or fragile. The dominant routing priority is energy efficiency: minimising battery drain to maximise the number of deliveries possible on a single charge cycle.

No existing UAV planner can switch between these three routing strategies based on a payload tag. The proposed two-tier system accomplishes this without any expert intervention or replanning. Payload metadata (fragility class, payload weight

class, delivery urgency) has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this project directly addresses.

1.3 Objectives

1. To formulate multi-objective UAV route planning as a structured **Pareto front pre-computation** problem over a delivery airspace graph, theoretically distinguishing the deterministic graph-search approach from stochastic population-based alternatives.

2. To design a **payload metadata schema** and a principled rule-based mapping that translates mission context (fragility class, delivery urgency, payload weight class) into Pareto point selection objective weights without any expert weight tuning.

3. To benchmark Tier 1 route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, measuring hypervolume, Pareto spread, and Inverted Generational Distance (IGD) as three distinct standard multi-objective quality indicators.

4. To quantify **Tier 2 selection accuracy** by measuring alignment between selected route characteristics and payload tag requirements across diverse mission types (fragile medical sample, emergency medicine, disaster-relief supplies).

5. To establish the system's **novelty**: payload metadata is used as the sole input to Pareto point selection, replacing context-blind entropy and diversity-based criteria with a deterministic algebraic mapping from payload description to route choice.

1.4 Scope

This project covers two phases. Phase I delivers a complete, validated software simulation system including the multi-objective A* graph-search engine (Tier 1), the payload-driven route selector (Tier 2), a MOPSO baseline, and a benchmarking suite evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes). Phase II extends the validated software to physical hardware: a Pixhawk-equipped F450 quadcopter with a Raspberry Pi 5 companion computer, validated through ArduPilot SITL simulation and outdoor flight tests. The Tier 1 Pareto front is treated as valid for a bounded refresh window rather than permanently static:

it is rebuilt on a periodic cadence, or when environmental drift (e.g. a substantial wind-condition shift) exceeds a defined threshold, and missions arriving within that window are served by Tier 2 against the current front without a per-mission recompute. The project does not address multi-UAV coordination, dynamic obstacle avoidance during flight, or real-time path replanning.

2. Related Work

Multi-objective UAV path planning sits at the intersection of graph-search and evolutionary optimisation [1]. Within it, route selection from a computed Pareto front is usually guided by the crowding-distance rule introduced by NSGA-II [4], or by an entropy criterion that keeps the archive diverse while a manually weighted scalar sum makes the final pick; neither considers delivery context. Payload information, where it appears at all, is limited to continuous mass folded directly into the energy/speed model itself, not discrete metadata used to select among a pre-computed Pareto front. Three papers most directly relevant to the proposed work are reviewed below; Table 2.1 gives a structured comparison.

Table 2.1: Summary of related work

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Chokkalingam & Ganesan (2026) [2]	Pareto-optimal particle swarm optimisation (POPSO) with dynamic parameter adaptation and Pareto–entropy archive selection to maintain front diversity during optimisation; the final single route is then read off via a weighted-sum scalarisation whose weights are set empirically per mission.	Entropy-guided archiving improves front diversity and adaptability to real-time environmental changes over static-weight baselines.	Final route choice still reduces to a manually re-tuned weighted sum per mission; payload metadata never incorporated, so fragile samples and emergency medicine receive the same hand-tuned selection procedure from the same front.

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Table 2.1 (continued)

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
2	Tsoutsouras et al. (2020) [3]	Adapts flight speed and the energy-consumption model to the UAV's instantaneous payload mass during trajectory planning for multi-user delivery missions, updating the trajectory as mass changes between drop-offs.	Mass-aware trajectory adaptation reduces energy consumption relative to mass-agnostic planning across multi-stop delivery routes.	Uses continuous payload mass only as a physical parameter inside the energy/speed model itself; builds no multi-objective Pareto front and has no notion of discrete payload metadata (fragility class, payload-weight class, urgency) selecting among pre-computed routing trade-offs.
3	Ren et al. (2019) [5]	Crowding-distance-based NSGA-II (CDNSGA-II) for multi-objective UAV path planning, balancing distance and safety over an octree-represented urban environment.	CDNSGA-II finds well-spread, collision-free Pareto-optimal paths efficiently in octree-modelled urban airspace.	Crowding distance encodes only the geometric spread of the front; it cannot represent urgency, fragility, or payload weight, and mission context plays no role in selection.

2.1 Research Gap

Two of the three reviewed papers treat Pareto-front diversity as the central design concern, and only one of those two uses it to *select* a route: the crowding distance of [5] picks a single point directly from the front, while [2] uses entropy only to

keep the archive *diverse* during optimisation and falls back to a manually weighted scalar sum, re-tuned per mission by a human operator, to make the actual final pick. [3] does not build a Pareto front at all: it folds continuous payload mass into the energy model's equations of motion and adapts flight speed as mass changes across a multi-drop mission, but never selects among competing routing trade-offs based on what is being carried. Whichever direction is taken, geometric diversity or continuous mass-driven physics adaptation, none of the three use payload weight, or any other payload property, as a discrete input to a Pareto-point *selection* mechanism; in [3], mass reshapes the cost model itself and never enters a selection formula. This is the structural gap the proposed system directly addresses: Tier 2 treats payload weight as one of four ordinal metadata tags algebraically blended into a selection weight, not a continuous physical parameter that reshapes the underlying cost model.

The gap runs deeper than an implementation detail: it is a design assumption baked into all three approaches. Entropy and crowding distance are both defined purely over the geometry of the Pareto front; mass-aware trajectory adaptation is defined purely over the physical energy model, with mass entering as a continuous parameter to that model, never as a selection criterion. None of these mechanisms has an interface through which discrete payload metadata, as opposed to continuous mass or point geometry, can enter a *selection* step. Extending them to incorporate payload information would require replacing the selection criterion entirely, which is precisely what the proposed Tier 2 utility scoring mechanism does.

Three specific gaps addressed by this work are as follows. **Gap 1 (No payload-aware Pareto selection):** Existing UAV multi-objective planners either apply a fixed geometric selection criterion independent of payload type, or offer no selection mechanism at all. The proposed Tier 2 introduces a formula-driven mapping from four payload tags directly to Pareto point selection. **Gap 2 (No re-use across mission types):** Existing planners re-run the full multi-objective search when the mission context changes. The proposed two-tier separation decouples Pareto front construction from mission arrivals: Tier 1 refreshes the front on a periodic cadence, independent of payload, and Tier 2 performs instantaneous route selection for any new payload tag against the current front, without re-running the search for each new mission. **Gap 3 (No algebraic bridge between delivery context and routing objectives):** The mapping between a payload description and the routing trade-off it implies (e.g., fragility $\rightarrow$ minimise risk, urgency $\rightarrow$ minimise distance) has never been formalised algebraically. The proposed weight-derivation formula provides this bridge, making the mapping deterministic, transparent, and reproducible.

3. Problem Statement

Existing UAV route planners either apply a fixed geometric selection criterion over the Pareto front, independent of payload type (Gap 1), or fold objectives into a **single hand-tuned weighted-sum scalar** before path search begins, permanently fixing the objective trade-off regardless of what is being carried. When the payload context changes, from fragile diagnostic samples requiring vibration-minimal routing to time-critical emergency medicine requiring shortest-path routing, an expert must manually re-tune the planner weights and re-run the search from scratch (Gap 2), because no algebraic mapping from payload description to routing trade-off exists to make that re-tuning unnecessary (Gap 3). This intervention is **infeasible in field conditions**.

Formally, given: (i) an airspace graph $G = (V, E)$ with three-objective edge cost vectors $\mathbf{c}(e) = [d(e), \varepsilon(e), \rho(e)]$ representing geodesic distance, estimated energy consumption, and flight risk respectively; (ii) a Pareto-optimal front $\mathcal{P}$ of non-dominated routes pre-computed over G; and (iii) a payload metadata tuple $(F, W, U, P_{\text{lvl}})$ encoding fragility class $F \in \{1 \dots 5\}$, payload weight class $W \in \{1 \dots 3\}$, delivery urgency $U \in \{1 \dots 3\}$, and priority level $P_{\text{lvl}} \in \{1 \dots 5\}$, the problem is to select the route $p^* \in \mathcal{P}$ that best satisfies the routing requirements implied by the payload metadata, without expert re-tuning and without re-running the multi-objective search.

The inputs are measurable quantities the operator can assign from domain knowledge (e.g., "this vial is fragile: $F = 5$; delivery is non-urgent: $U = 1$"). The constraints are that selection must be deterministic, reproducible, and complete within one second of payload tag entry. The success criterion is that the selected route characteristics (distance, energy, risk) align with the payload's stated requirements as verified by a Tier 2 selection accuracy metric evaluated across multiple payload presets. Closing Gap 1 requires p^* to depend on $(F, W, U, P_{\text{lvl}})$ rather than on front geometry alone; closing Gap 2 requires $\mathcal{P}$ to be reused across missions between scheduled refreshes rather than rebuilt for every mission; closing Gap 3 requires the map from $(F, W, U, P_{\text{lvl}})$ to p^* to be an explicit algebraic formula rather than an expert-set rule or lookup table.

4. Justification for the Problem Formulation

4.1 Need for the Proposed Work

The gap established in Chapter 2 carries a direct operational consequence. Manually re-tuning three objective weights and re-running a multi-objective search between every dispatch requires an expert on hand at the point of launch, which is unavailable in a time-critical emergency deployment. A system that requires this intervention is unusable in the field regardless of its routing quality on paper.

This is what motivates separating Pareto front construction from payload-driven selection. The front is built offline for a delivery region and refreshed on a periodic cadence, or when environmental drift (e.g. a substantial wind-condition shift) exceeds a defined threshold, rather than rebuilt for every mission; each mission arriving within that refresh window runs only Tier 2, reading the payload tag and selecting a route from the current front in near-instantaneous time. No expert, no per-mission re-planning, no delay between payload tagging and mission file generation.

4.2 Feasibility

Technical feasibility: The Tier 1 multi-objective A*/Dijkstra algorithm is a well-studied graph-search extension. The Tier 2 utility scoring is a closed-form algebraic formula executable in microseconds on any hardware. Both components are implementable in Python with standard open-source libraries (NetworkX [6], NumPy). Hardware deployment on a Raspberry Pi 5 companion computer is feasible given the computational simplicity of Tier 2 route selection ($< 1\,\text{s}$ end-to-end from payload tag input to mission file output).

Operational feasibility: The system requires only four integer tag inputs from the operator, eliminating the need for weight-tuning expertise. The ArduPilot SITL simulation environment allows complete end-to-end validation before outdoor flights, reducing hardware risk to an acceptable level.

Economic feasibility: All software components use open-source libraries and protocols (osmnx [7], NetworkX [6], NumPy, matplotlib, MAVLink [8]). The hardware platform (F450 quadcopter frame, Pixhawk flight controller, Raspberry Pi 5) uses standard, commercially available components at modest cost. No specialised or proprietary software licences are required.

Schedule feasibility: Phase I (software simulation and benchmarking) is achievable within the first semester aligned to Review I. Phase II (hardware integration

and outdoor testing) is planned for the second semester, with milestones aligned to the Review II and Final Review schedule as shown in Chapter 6.

4.3 Expected Contribution

Pareto-optimal graph-search is an established technique; the contribution here is not the search itself, but the fact that the bridge connecting Tier 1's routing objectives to Tier 2's payload metadata is algebraic rather than a lookup table or an expert-set rule. Because every weight is a function of all three raw scores simultaneously (Chapter 5), the resulting utility function can select any point on the front, not only its extremes, and does so without a human deciding which extreme applies to which payload. No prior UAV delivery system grants payload metadata this role: existing planners treat route selection as a front-diversity problem, blind to what is being carried. In practice, a periodically refreshed front serves every mission profile a delivery region sees between refreshes, and route choice becomes a sub-second metadata lookup.

5. Methodology

The system rests on one architectural decision: separate Pareto front construction (Tier 1) from route selection (Tier 2), for the reasons given in Chapter 4. The front is built per delivery region and refreshed on a periodic cadence rather than per mission; each mission arriving within a refresh window runs only Tier 2, reading the payload tag and selecting a route from the current front in near-instantaneous time.

5.1 Architectural Design for Proposed System

Figure 5.1 shows the complete two-tier system architecture. Airspace graph data from OpenStreetMap enters Tier 1, which performs a multi-objective A*/Dijkstra search and stores the resulting Pareto-optimal route set. For each new mission, Tier 2 reads the four operator-assigned payload tags, derives objective weights algebraically, scores every stored Pareto point via a utility function, and outputs the arg max as the selected mission route along with an ArduPilot-compatible `.waypoints` file.

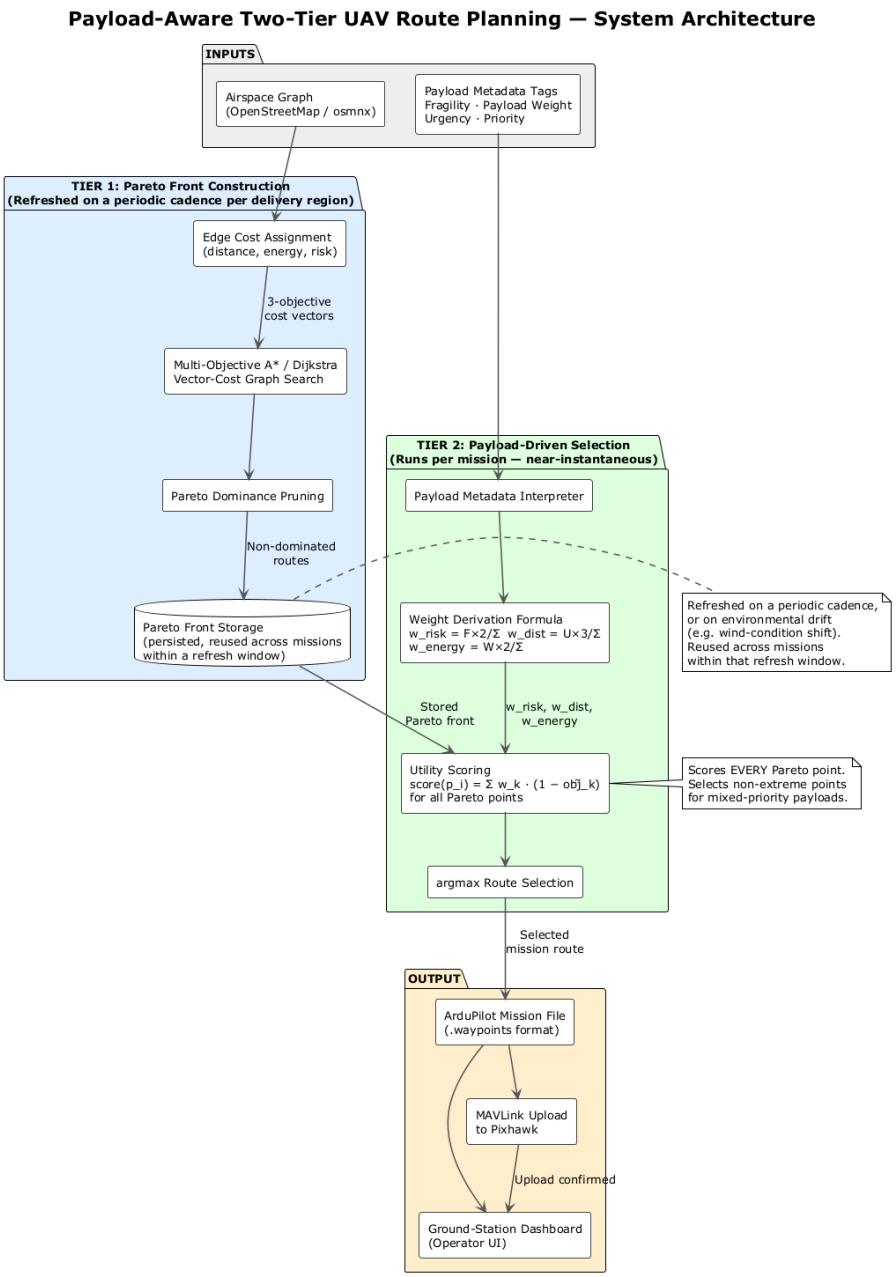


Figure 5.1: Overall system architecture: Two-Tier Payload-Aware UAV Route Planner. Tier 1 builds the Pareto front per delivery region on a periodic refresh cadence; Tier 2 selects the optimal route for each new payload tag using the current front.

The airspace graph $G = (V, E)$ is extracted from OpenStreetMap using the `osmnx` Python library [7]. Each edge $e \in E$ carries a cost vector:

$$\mathbf{c}(e) = \Big[d(e),\ \varepsilon(e),\ \rho(e)\Big]$$

where $d(e)$ is the geodesic distance of the flight segment (km), $\varepsilon(e)$ is the estimated energy consumption (proxy: $d(e)\times$ altitude-change factor), and $\rho(e) \in [0, 1]$ is the flight risk score derived from obstacle density, proximity to no-fly zones, and wind

exposure (simulated programmatically for reproducible benchmarking).

The three optimisation objectives are: (1) minimise route distance d; (2) minimise energy consumption ε; and (3) minimise flight risk ρ. A route p dominates route q if p is at least as good as q on all three objectives and strictly better on at least one. The Pareto front $\mathcal{P}$ is the set of all non-dominated routes.

Tier 2 Algebraic Bridge. Each of the four payload metadata tags maps to exactly one Tier 1 routing objective as shown in Table 5.1.

Table 5.1: Semantic mapping: payload metadata tags to Tier 1 routing objectives

Payload Tag	**Range**	**Maps to Tier 1 Objective**	**Reasoning**
Fragility class F	1–5	w_{risk} (flight risk weight)	Fragile items need low-vibration, low-risk corridors
Payload weight class W	1–3	w_{energy} (energy weight)	Heavier payload increases hover/lift power draw ⇒ higher energy cost
Delivery urgency U	1–3	w_{dist} (distance weight)	Time-critical deliveries need the shortest path
Priority level P_{lvl}	1–5	Tie-breaker amplifier	Scales the dominant weight when scores are close

From the payload tags, three routing objective weights are derived algorithmically with no expert input:

$$r_{\text{raw}} = F \times 2, \quad d_{\text{raw}} = U \times 3, \quad e_{\text{raw}} = W \times 2 \tag{5.1}$$

$$w_{\text{risk}} = \frac{r_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{dist}} = \frac{d_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{energy}} = \frac{e_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{5.2}$$

The multipliers (2 for fragility and payload weight, 3 for urgency) reflect the domain insight that urgency has a higher penalty sensitivity than fragility or payload weight in the UAV medical delivery context. Payload weight is bound to the energy objective because heavier payload directly increases hover/lift power draw, a physically grounded relationship absent for risk or distance.

For each Pareto point p_i with objective values $(d_i, \varepsilon_i, \rho_i)$, min-max normalised

values $\tilde{d}_i$, $\tilde{\varepsilon}_i$, $\tilde{\rho}_i$ are computed over the full Pareto front $\mathcal{P}$. The utility score is:

$$\text{score}(p_i) = w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i) \tag{5.3}$$

The selected mission route is $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$. For high-priority missions ($P_{\text{lvl}} \geq 4$), the dominant weight is amplified by $0.1 \times w_{\max}$ to sharpen the selection.

The Tier 1 and Tier 2 activity flows are shown in Figures 5.2 and 5.3 respectively.

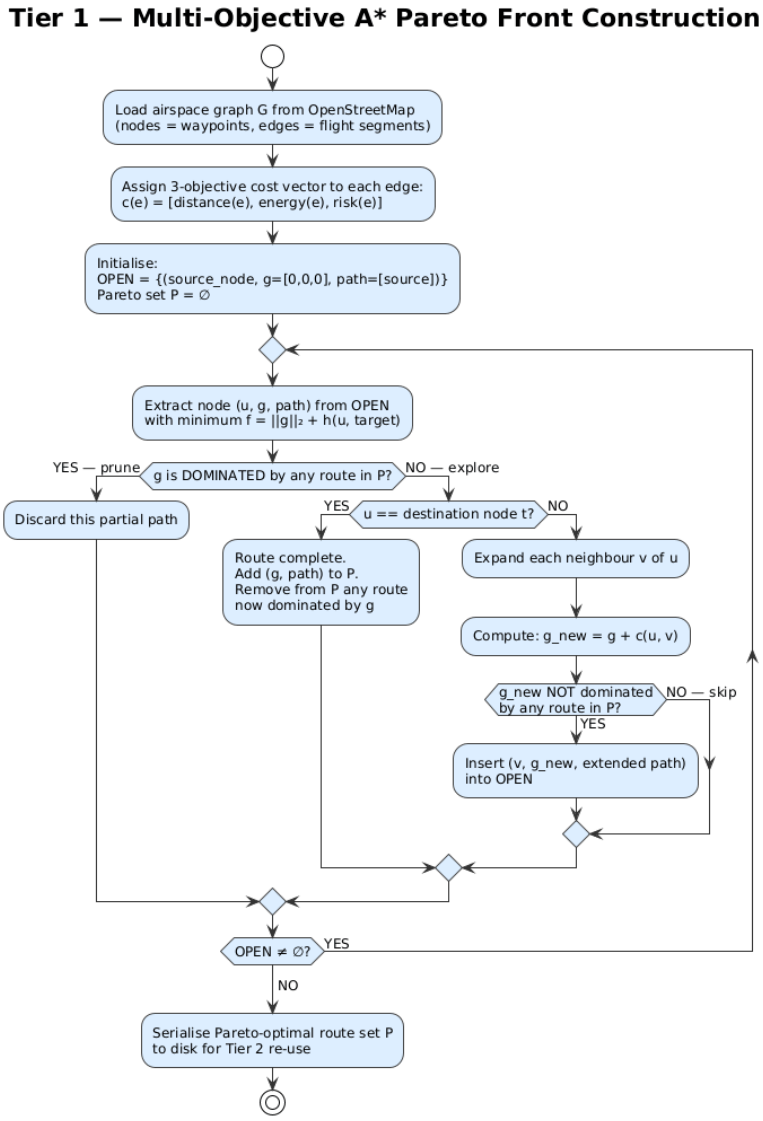


Figure 5.2: Tier 1 activity flow: Multi-Objective A* Pareto Front Construction. The open set is explored in order of estimated total cost until no further non-dominated routes exist.

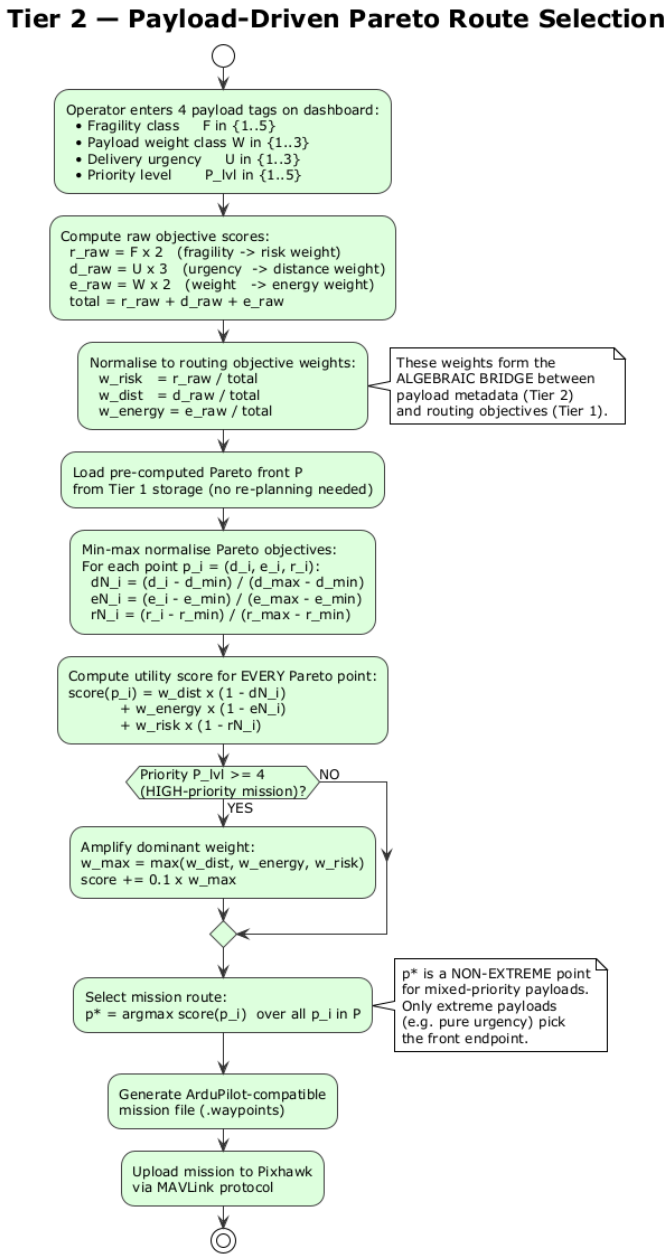


Figure 5.3: Tier 2 activity flow: Payload-Driven Pareto Selection. Payload tags are algebraically mapped to weights, which score every Pareto point via a utility function; the arg max is selected as the mission route.

5.2 Modules

5.2.1 Module 1: Airspace Graph Builder

Extracts the road/path network from OpenStreetMap for a specified delivery region using `osmnx`. Assigns three objective costs to each edge: geodesic distance, energy (distance × altitude-change factor), and flight risk (simulated threat zones and obstacle

density scored 0–1 using a configurable hazard map). Output: a directed weighted graph G stored as a serialised NetworkX object ready for Tier 1 search.

5.2.2 Module 2: Multi-Objective A*/Dijkstra Engine (Tier 1)

Implements the multi-objective A* algorithm maintaining a cost vector $\mathbf{g}$ (not a scalar) for each partial path. Paths dominated by already-discovered routes are pruned early. The `UpdateParetoFront` subroutine adds a new route and removes any previously stored routes it dominates. A node-level label-dominance check (NAMOA*) additionally discards any partial path whose cost vector is already dominated by a previously seen label at the same node, before it enters the open set, which avoids the exponential blow-up of naive MO-A* on dense graphs. Output: a deterministic, reproducible Pareto front $\mathcal{P}$ serialised to disk for Tier 2 re-use across all future missions in the same delivery region.

Algorithm 1 NAMOA* – Multi-Objective A*/Dijkstra Pareto Front Construction (Tier 1)

Require: Airspace graph $G = (V, E)$, source s, target t, edge cost $c(u, v) = [d, \varepsilon, \rho]$
Ensure: Pareto-optimal route set $\mathcal{P}$

$\mathcal{P} \leftarrow \emptyset$; $\quad labels[v] \leftarrow \emptyset$ for all $v \in V$
$g_0 \leftarrow [0, 0, 0]$; $\quad$ push $(f(g_0, s), g_0, s, \{s\})$ onto priority queue *Open*
$labels[s] \leftarrow \{g_0\}$
while *Open* $\neq \emptyset$ **do**
  $(f, g, u, visited) \leftarrow$ PopMin(*Open*)
  if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g$ **then** ▷ dominated by global front
    continue
  end if
  if $u = t$ **then**
    UpdateParetoFront($\mathcal{P}, g, \text{path}(u)$)
    continue
  end if
  for each successor v of u with $v \notin visited$ **do**
    $g_{\text{new}} \leftarrow g + c(u, v)$
    if $\exists\, \ell \in labels[v]$ such that $\ell \prec g_{\text{new}}$ **then**
      continue ▷ node-level label dominance – NAMOA* pruning
    end if
    $labels[v] \leftarrow \{\ell \in labels[v] : g_{\text{new}} \not\prec \ell\} \cup \{g_{\text{new}}\}$
    if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g_{\text{new}}$ **then**
      continue
    end if
    $f_{\text{new}} \leftarrow \|g_{\text{new}}\| + h(v, t)$
    push $(f_{\text{new}}, g_{\text{new}}, v, visited \cup \{v\})$ onto *Open*
  end for
end while
return $\mathcal{P}$

procedure UpdateParetoFront($\mathcal{P}, g_{\text{new}}, \text{path}_{\text{new}}$)
  if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g_{\text{new}}$ **then return false**
  end if
  $\mathcal{P} \leftarrow \{(g, p) \in \mathcal{P} : g_{\text{new}} \not\prec g\}$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{(g_{\text{new}}, \text{path}_{\text{new}})\}$
  return true
end procedure

5.2.3 Module 3: Payload Metadata Interpreter (Tier 2)

Implements the payload-driven route selection algorithm: reads the four payload tags, derives objective weights via Equations (5.1)–(5.2), applies min-max normalisation to Pareto objectives, computes a utility score for every Pareto point via Equation (5.3), and returns the arg max route. Optionally amplifies the dominant weight for high-priority missions ($P_{\text{lvl}} \geq 4$). Generates an ArduPilot-compatible `.waypoints` mission file for upload.

Algorithm 2 Payload-Driven Pareto Route Selection (Tier 2)

Require: Pareto front $\mathcal{P} = \{(d_i, \varepsilon_i, \rho_i)\}$, payload tags F, W, U, P_{lvl}
Ensure: Selected mission route p^*

$r_{\text{raw}} \leftarrow F \times 2$; $\quad d_{\text{raw}} \leftarrow U \times 3$; $\quad e_{\text{raw}} \leftarrow W \times 2$
$total \leftarrow r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}$
$w_{\text{risk}} \leftarrow r_{\text{raw}}/total$; $\quad w_{\text{dist}} \leftarrow d_{\text{raw}}/total$; $\quad w_{\text{energy}} \leftarrow e_{\text{raw}}/total$
if $P_{\text{lvl}} \geq 4$ **then** ▷ high-priority amplification
    $w_{\max} \leftarrow \max(w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}})$
    add $0.1 \times w_{\max}$ to whichever of $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ equals $w_{\max}$
    renormalise $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ so they sum to 1
end if
compute $d_{\min}, d_{\max}, \varepsilon_{\min}, \varepsilon_{\max}, \rho_{\min}, \rho_{\max}$ over $\mathcal{P}$
for each $p_i = (d_i, \varepsilon_i, \rho_i) \in \mathcal{P}$ **do**
    $\tilde{d}_i \leftarrow (d_i - d_{\min})/(d_{\max} - d_{\min})$ ▷ 0 if $d_{\max} = d_{\min}$
    $\tilde{\varepsilon}_i \leftarrow (\varepsilon_i - \varepsilon_{\min})/(\varepsilon_{\max} - \varepsilon_{\min})$
    $\tilde{\rho}_i \leftarrow (\rho_i - \rho_{\min})/(\rho_{\max} - \rho_{\min})$
    $\text{score}(p_i) \leftarrow w_{\text{dist}}(1 - \tilde{d}_i) + w_{\text{energy}}(1 - \tilde{\varepsilon}_i) + w_{\text{risk}}(1 - \tilde{\rho}_i)$
end for
$p^* \leftarrow \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$
return p^*

5.2.4 Module 4: Testing and Evaluation

Evaluates all methods on three scenario scales: S1 (≈50 nodes, low obstacle density), S2 (≈200 nodes, medium density), and S3 (≈500 nodes, high density). The MOPSO baseline [9] runs with swarm size 100, 200 iterations, and a Pareto archive of 100 solutions, repeated 30 times per scenario to obtain mean ± standard deviation. Metrics include Hypervolume (HV), Pareto spread (the standard Δ metric, capturing extent and uniformity of the front), Inverted Generational Distance (IGD) computed against a reference front formed from the union of all non-dominated solutions found across

every method and repetition for each scenario, and Tier 2 selection accuracy across four payload presets. Phase II testing extends to ArduPilot SITL validation and outdoor flight tests on an F450 quadcopter with Pixhawk flight controller, measuring route adherence (GPS position error vs. planned route) and end-to-end latency.

6. Project Timeline Chart

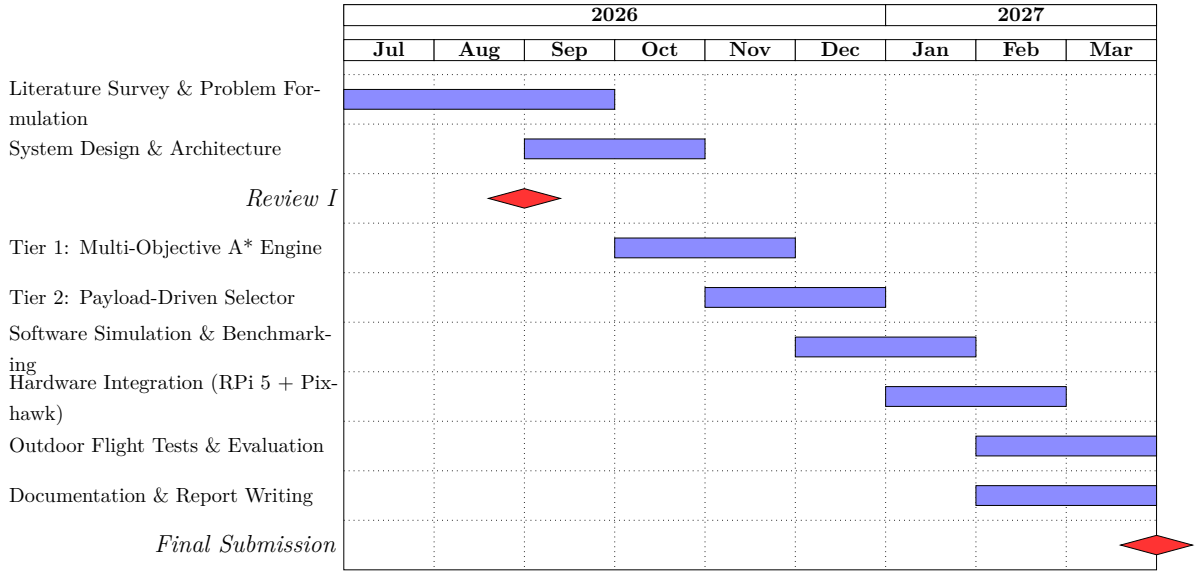


Figure 6.1: Project timeline (July 2026 – March 2027). Red diamonds mark review and submission milestones.

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